

COLORADO STATE UNIVERSITY PUEBLO

School of Engineering

A Graph Neural Surrogate for Longitudinal Train Dynamics

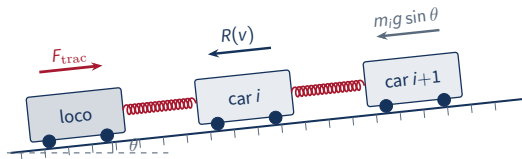
Toward Fuel-Optimal Control of Freight Consists

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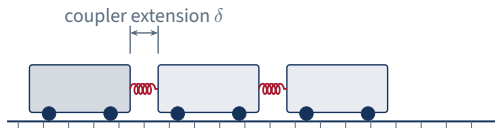
Background: What Longitudinal Train Dynamics Are

- A freight train is a **distributed** mechanical system: tens to hundreds of coupled masses.
- Each vehicle samples its **own** grade, curvature and resistance along the route.
- Couplers carry **slack**: free travel before any force is transmitted.
- Quantities of operational interest: **fuel**, **arrival time**, **peak coupler force**.



$$m_i \dot{v}_i = F_{\text{ext},i} + F_{i-1} - F_i$$

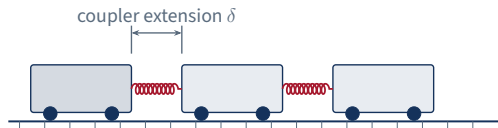
Slack Is the Whole Problem



bunched — slack closed, no force transmitted



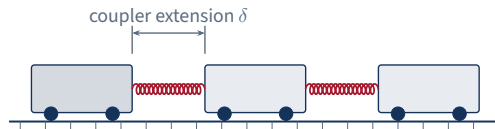
Slack Is the Whole Problem



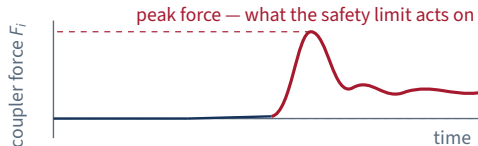
slack taken up — couplers reach the deadband edge



Slack Is the Whole Problem



run-out — force spikes the instant slack closes



A run-in/run-out event is what breaks a knuckle — and the deadband is a **derivative discontinuity** that forces an adaptive solver to collapse its step size.